NexRay AI

Medical Assistant Platform • Clinical Decision Support Report

10 August 2026 | 07:21 PM

Session ID: 49 **Modules Used:** Symptom Checker **Date:** 10 August 2026 | 07:21 PM

Symptom Analysis

Possible Condition	Confidence
Cholera	92%
Acute Gastroenteritis (Viral or Bacterial)	68%
Typhoid Fever with Severe Gastroenteritis	35%

■ **Urgency Level: Emergency — Immediate action required**

Recommended Tests

- Stool culture and sensitivity on selective media (TCBS agar for Vibrio cholerae)
- Stool microscopy and rapid cholera antigen test if available
- Serum electrolytes (sodium, potassium, chloride, bicarbonate) to assess dehydration and acid-base status
- Blood culture to rule out bacteremia
- Rapid diagnostic test for rotavirus/norovirus if available

Suggested Treatment

- Immediate aggressive oral rehydration therapy (ORT) with WHO low-osmolarity solution containing sodium chloride 2.6g/L, potassium chloride 1.5g/L, glucose 13.5g/L, and trisodium citrate dihydrate 2.9g/L
- Intravenous fluid replacement if unable to tolerate oral fluids or signs of severe dehydration - use isotonic saline or Ringer's lactate
- Azithromycin 10mg/kg single dose OR doxycycline 2-4mg/kg divided doses (if age permits) as first-line antibiotic for confirmed cholera
- Zinc supplementation 20mg daily for 10-14 days to reduce duration and severity of diarrhea
- Continue breastfeeding or age-appropriate nutrition once vomiting subsides

Next Steps

- Admit to hospital for close monitoring of fluid balance, electrolytes, and vital signs - high risk of hypovolemic shock in children
- Establish IV access and begin fluid resuscitation immediately based on degree of dehydration (assess by skin turgor, capillary refill, urine output)
- Monitor urine output, ongoing stool losses, and reassess hydration status hourly during acute phase
- Send stool samples for culture and sensitivity before starting antibiotics if possible
- Isolate patient and implement standard and contact precautions; report suspected cholera to public health authorities as it is a notifiable disease
- Educate caregivers on handwashing and safe water use to prevent secondary transmission

Summary: *An 11-year-old female presenting with classic signs of acute cholera including rice-water stools, severe dehydration, vomiting, and electrolyte loss within 12 hours. This is a medical emergency requiring immediate hospitalization, aggressive fluid and electrolyte replacement, and antimicrobial therapy. Cholera has 92% confidence based on the pathognomonic presentation. Immediate intervention is critical to prevent hypovolemic shock and death in this age group. Public health notification is required.*

■ ***DISCLAIMER:*** *This report is generated by NexRay AI and is intended solely as a clinical decision-support tool. All findings, conditions, treatments, and recommendations are AI-generated suggestions and do not constitute a formal medical diagnosis or prescription. The clinical judgment of the attending medical professional must be applied before any action is taken.*